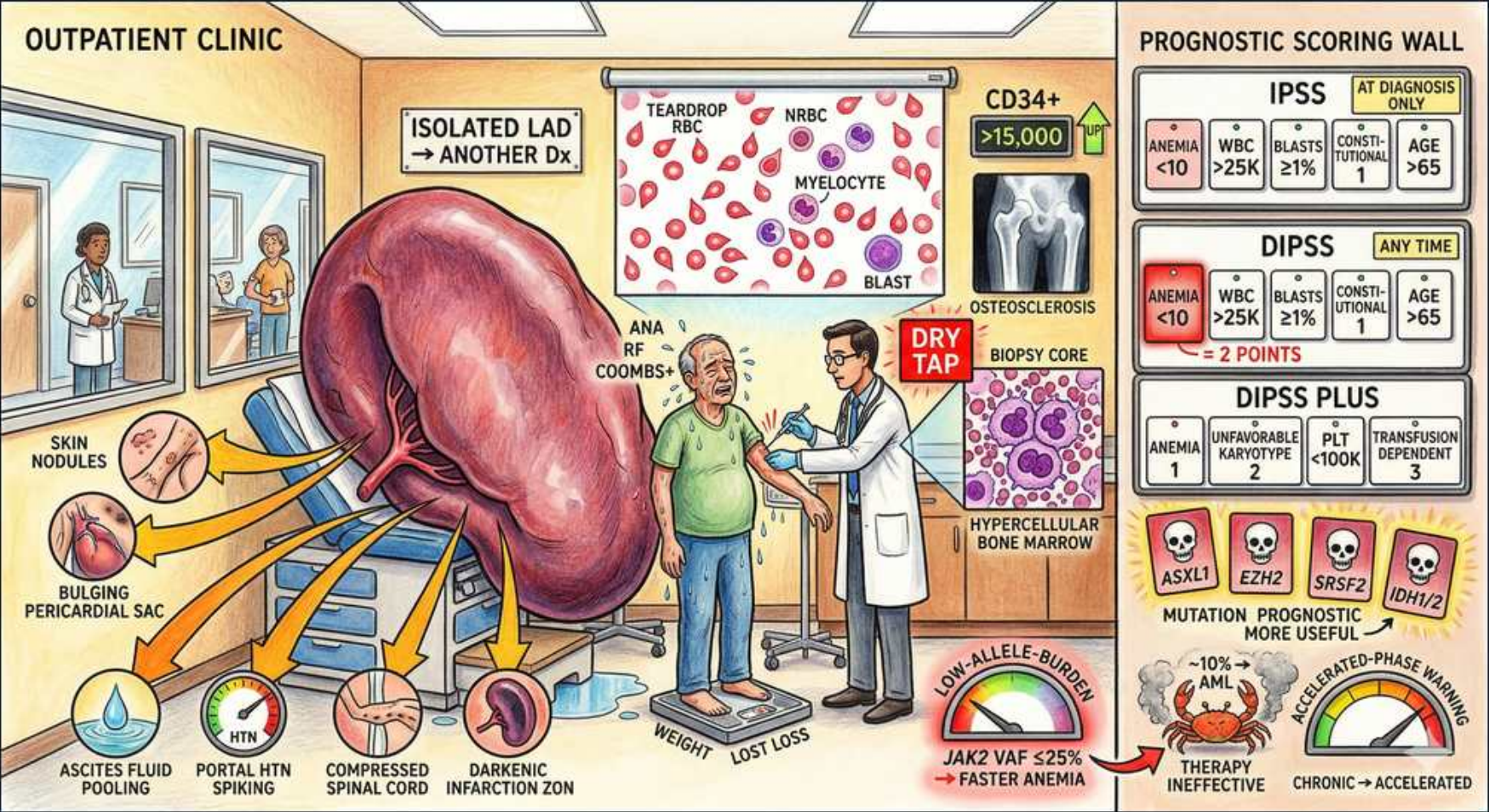


Polycythemia / MPN — Myelofibrosis clinical features & diagnosis



1. Teardrop RBCs (dacrocytes)

WHAT YOU SEE

TEARDROP RBC blood smear panel top

WHAT IT ENCODES

Dacrocytes (teardrop-shaped RBCs) are the classic peripheral-smear hallmark of myelofibrosis — red cells deformed as they squeeze through the fibrotic marrow and through splenic cords during extramedullary hematopoiesis. On boards, teardrop cells + nucleated RBCs + immature myeloid cells = marrow infiltration/fibrosis.

BOARD TRAP

WRONG: dacrocytes are specific for primary myelofibrosis. They are NOT — teardrop cells appear in ANY myelophthisic/infiltrative marrow process (metastatic cancer, thalassemia, myelodysplasia), so they indicate marrow disruption, not a specific diagnosis.

2. Leukoerythroblastic smear

WHAT YOU SEE

NRBC, MYELOCYTE, BLAST labeled on smear

WHAT IT ENCODES

A leukoerythroblastic blood picture = nucleated RBCs (normoblasts) PLUS immature myeloid precursors (myelocytes, metamyelocytes, occasional blasts) spilling into the peripheral blood. It reflects marrow stress/infiltration forcing premature release and extramedullary hematopoiesis. Pair with teardrop cells for the myelofibrosis pattern.

BOARD TRAP

WRONG: a leukoerythroblastic smear with a few circulating blasts means the patient already has AML. A small number of circulating blasts is expected in chronic-phase PMF; AML transformation requires >=20% blasts in blood or marrow.

3. Dry tap (fibrotic marrow)

WHAT YOU SEE

DRY TAP syringe label

WHAT IT ENCODES

Bone marrow ASPIRATION fails to yield a sample ('dry tap' / punctio sicca') because the dense reticulin/collagen fibrosis prevents marrow from being drawn up. Diagnosis therefore requires a marrow BIOPSY (core), not aspirate, to demonstrate fibrosis and megakaryocyte atypia.

BOARD TRAP

WRONG: when the aspirate is dry you can still establish the diagnosis from the aspirate smear alone. You cannot — a dry tap mandates a trephine BIOPSY; relying on the failed aspirate misses the fibrosis and atypical megakaryocyte clusters that define PMF.

4. Hypercellular fibrotic marrow

WHAT YOU SEE

BIOPSY CORE HYPERCELLULAR BONE MARROW

WHAT IT ENCODES

Early/cellular-phase PMF biopsy shows a HYPERcellular marrow with clustered, atypical (bulbous, hyperchromatic) megakaryocytes and increasing reticulin fibrosis; later 'fibrotic phase' becomes hypocellular with dense collagen and osteosclerosis. Reticulin stain grades fibrosis MF-0 to MF-3.

BOARD TRAP

WRONG: a hypercellular marrow rules out myelofibrosis (which should look empty/fibrotic). Early PMF is typically HYPERcellular — the hypocellular, scarred marrow is a late finding, so cellularity does not exclude the diagnosis.

5. Massive splenomegaly

WHAT YOU SEE

Huge red spleen dominating center-left

WHAT IT ENCODES

Massive splenomegaly (spleen often crossing the midline/into the pelvis) is the most prominent physical finding in PMF, driven by extramedullary hematopoiesis plus portal congestion. It causes early satiety, LUQ pain, and can infarct. Differential for MASSIVE splenomegaly: PMF, CML, hairy cell leukemia, malaria, visceral leishmaniasis, Gaucher.

BOARD TRAP

WRONG: massive splenomegaly in an older adult points first to lymphoma. The classic causes of MASSIVE (rather than mild) splenomegaly are myeloproliferative — PMF and CML — plus chronic infections/storage disease; most lymphomas give only mild-moderate enlargement.

6. Extramedullary hematopoiesis

WHAT YOU SEE

Arrows to skin nodules, pericardial sac, portal HTN, cord

WHAT IT ENCODES

With marrow displaced by fibrosis, hematopoiesis relocates outside the marrow (extramedullary hematopoiesis, EMH). Beyond spleen/liver, EMH foci can appear in skin, lymph nodes, pleura/pericardium (effusions/tamponade), paraspinal tissue (spinal cord compression — a radiotherapy emergency), and cause portal hypertension/ascites.

BOARD TRAP

WRONG: EMH is harmless residual blood production. EMH foci can be clinically dangerous — paraspinal EMH causing cord compression and pericardial/pleural EMH effusions are emergencies treated with low-dose radiation.

7. Osteosclerosis on X-ray

WHAT YOU SEE

X-ray with OSTEOSCLEROSIS label

WHAT IT ENCODES

Plain radiographs in advanced PMF may show OSTEOSCLEROSIS — diffusely increased bone density, especially axial skeleton and metaphyses — reflecting the osteoblastic/OPG imbalance accompanying marrow fibrosis.

BOARD TRAP

WRONG: bony lesions in this MPN are lytic. PMF produces sclerotic (dense) bone, not lytic lesions; lytic 'punched-out' lesions point to multiple myeloma instead.

8. Constitutional symptoms

WHAT YOU SEE

WEIGHT LOSS scale, exhausted patient

WHAT IT ENCODES

A heavy inflammatory cytokine burden (TNF-alpha, IL-6, IL-8) drives constitutional symptoms: profound fatigue, weight loss, drenching night sweats, low-grade fevers, pruritus, and cachexia. Symptom burden is captured by the MPN-SAS/MPN10 score and is a target of JAK-inhibitor therapy.

BOARD TRAP

WRONG: constitutional symptoms in PMF always signal leukemic transformation and need a marrow biopsy first. They are usually cytokine-driven features of chronic-phase disease itself; transformation is defined by rising blasts, not symptoms alone.

9. IPSS at diagnosis

WHAT YOU SEE

IPSS box: anemia, WBC>25K, blasts, constitutional, age>65

WHAT IT ENCODES

The IPSS (International Prognostic Scoring System) is applied ONLY at diagnosis. Its 5 adverse factors, each 1 point: hemoglobin <10 g/dL, WBC >25,000/uL, peripheral blasts >=1%, constitutional symptoms, and age >65. Risk strata range from low (0) to high (>=3) and guide whether to pursue transplant.

BOARD TRAP

WRONG: IPSS can be reapplied during follow-up to track changing risk. IPSS is a baseline-only (at-diagnosis) tool; for risk at any later time point you must use DIPSS or DIPSS-Plus.

10. DIPSS (any time)

WHAT YOU SEE

DIPSS box, anemia = 2 points

WHAT IT ENCODES

The Dynamic IPSS (DIPSS) uses the same variables but can be applied at ANY time during the disease course, and it weights anemia (Hgb <10) as 2 points (the others 1 point each). This lets clinicians recompute prognosis as the patient evolves.

BOARD TRAP

WRONG: DIPSS and IPSS score anemia identically. The key difference is that DIPSS gives anemia DOUBLE weight (2 points) and, unlike IPSS, is valid at any time, not just at diagnosis.

11. DIPSS Plus (karyotype/platelets)

WHAT YOU SEE

DIPSS PLUS: unfavorable karyotype, plt<100K, transfusion-dependent

WHAT IT ENCODES

DIPSS-Plus adds three more adverse factors to DIPSS: unfavorable karyotype, platelet count <100,000/uL, and red-cell transfusion dependence. It refines risk stratification and is the version most often cited for transplant decision-making.

BOARD TRAP

WRONG: DIPSS-Plus is a molecular score based on mutation panels. DIPSS-Plus is cytogenetic/clinical (karyotype, platelets, transfusion need); mutation-based refinement (ASXL1, etc.) comes from MIPSS70/GIPSS, not DIPSS-Plus.

12. High-risk mutations

WHAT YOU SEE

Skull mutations ASXL1, EZH2, SRSF2, IDH1/2

WHAT IT ENCODES

High-molecular-risk (HMR) mutations — ASXL1, EZH2, SRSF2, and IDH1/2 — independently predict worse survival and higher leukemic transformation; >=2 HMR mutations is especially adverse. These feed mutation-enhanced scores (MIPSS70/MIPSS70+, GIPSS) used in younger transplant candidates.

BOARD TRAP

WRONG: these high-risk mutations are the disease-defining DRIVER mutations of PMF. They are NOT drivers — the drivers are JAK2/CALR/MPL; ASXL1/EZH2/SRSF2/IDH1-2 are SUBCLONAL prognostic modifiers that worsen outcome.

13. Low JAK2 VAF → faster anemia

WHAT YOU SEE

LOW-ALLELE-BURDEN JAK2 VAF ≤25% → faster anemia

WHAT IT ENCODES

A LOW JAK2 V617F allele burden (variant allele frequency ≤25%) is paradoxically associated with more aggressive disease — faster development of anemia and worse outcomes — because a low driver burden often reflects a clone enriched for high-risk subclonal mutations or a more myelodepleted, fibrotic marrow.

BOARD TRAP

WRONG: higher JAK2 allele burden always means worse prognosis. In PMF the opposite can hold — a LOW JAK2 VAF flags more rapid anemia/progression, an exam-favorite counterintuitive point.

14. Accelerated → AML

WHAT YOU SEE

ACCELERATED-PHASE WARNING chronic→accelerated

WHAT IT ENCODES

PMF can progress from chronic phase to accelerated phase (10-19% blasts in blood/marrow) and then to secondary AML (>=20% blasts). Post-MPN AML is treatment-resistant with poor survival; rising blast count, worsening cytopenias, and new constitutional symptoms herald transformation.

BOARD TRAP

WRONG: any blasts in the blood of a PMF patient define AML transformation. Chronic-phase PMF normally has rare circulating blasts; ACCELERATED phase is 10-19% and AML requires >=20% blasts in blood or marrow.